```
# EXPERIMENT NO. 1
# IMPLEMENTATION OF FUNDAMENTAL IMAGE PROCESSING OPERATIONS
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os
import shutil
import pandas as pd
from google.colab import files
from IPython.display import display
print("=" * 75)
print("EXPERIMENT NO. 1")
print("Implementation of Fundamental Image Processing Operations")
print("using Python and OpenCV")
print("=" * 75)
# -------------------- 2. UPLOAD IMAGE ----------------------------
print("\n📂 Select a color image from your PC...")
uploaded = files.upload()
if len(uploaded) == 0:
    raise Exception("❌ No image uploaded.")
input_filename = list(uploaded.keys())[0]
print(f"\n✅ Image uploaded: {input_filename}")
# -------------------- 3. CREATE OUTPUT FOLDER -------------------
output_dir = "/content/Experiment_01_Results"
if os.path.exists(output_dir):
    shutil.rmtree(output_dir)
os.makedirs(output_dir)
# -------------------- 4. READ IMAGE ------------------------------
image_bgr = cv2.imread(input_filename)
if image_bgr is None:
    raise Exception("❌ Could not read the uploaded image.")
# Convert BGR to RGB for Matplotlib
```

```
)
plt.show()
# -------------------- 7. DISPLAY USING OPENCV ------------------
# OpenCV internally uses BGR.
# The image is converted to RGB only for correct Matplotlib display.
print("\n✅ Image successfully loaded using OpenCV.")
print("✅ OpenCV internally represents color images in BGR format.")
# -------------------- 8. SAVE JPEG AND PNG ----------------------
jpeg_path = os.path.join(
    output_dir,
    "02_Image_Saved_JPEG.jpg"
)
png_path = os.path.join(
    output_dir,
    "03_Image_Saved_PNG.png"
)
cv2.imwrite(
    jpeg_path,
    image_bgr,
    [cv2.IMWRITE_JPEG_QUALITY, 95]
)
cv2.imwrite(
    png_path,
    image_bgr
)
jpeg_size = os.path.getsize(jpeg_path)
png_size = os.path.getsize(png_path)
print("\n" + "=" * 70)
print("FILE FORMAT COMPARISON")
print("=" * 70)
print(
    f"JPEG File Size : {jpeg_size / 1024:.2f} KB"
)
```

```
    ),
    h_channel
)
cv2.imwrite(
    os.path.join(
        output_dir,
        "07_HSV_S_Channel.png"
    ),
    s_channel
)
cv2.imwrite(
    os.path.join(
        output_dir,
        "08_HSV_V_Channel.png"
    ),
    v_channel
)
# -------------------- 11. LAB COLOR SPACE ----------------------
lab = cv2.cvtColor(
    image_bgr,
    cv2.COLOR_BGR2LAB
)
l_channel, a_channel, b_channel = cv2.split(lab)
cv2.imwrite(
    os.path.join(
        output_dir,
        "09_LAB_Image.png"
    ),
    lab
)
cv2.imwrite(
    os.path.join(
        output_dir,
```

```
)
axes[1, 0].axis("off")
# LAB converted back to RGB for visualization
lab_display = cv2.cvtColor(
    lab,
    cv2.COLOR_LAB2RGB
)
axes[1, 1].imshow(lab_display)
axes[1, 1].set_title(
    "LAB Color Space",
    fontsize=16
)
axes[1, 1].axis("off")
plt.suptitle(
    "Color Space Conversion",
    fontsize=21
)
plt.tight_layout()
plt.savefig(
    os.path.join(
        output_dir,
        "13_Color_Space_Comparison.png"
    ),
    dpi=220,
    bbox_inches="tight"
)
plt.show()
# -------------------- 13. RESIZING ------------------------------
# Resize to 50% of original dimensions
new_width = max(1, width // 2)
new_height = max(1, height // 2)
resized = cv2.resize(
    image_bgr,
```

```
for filename in sorted(
    os.listdir(output_dir)):
print("✓", filename)
print("\n" + "=" * 75)
print("✅ EXPERIMENT NO. 1 COMPLETED SUCCESSFULLY")
print("=" * 75)
print(f"\n📁 All results are saved in:\n{output_dir}"
)
```

OUTPUT - 1

✅ Image uploaded: EXP_01_SAMP_IMG.jpg

IMAGE PROPERTIES

Image Name : EXP_01_SAMP_IMG.jpg

Width : 736 pixels

Height : 1308 pixels

Resolution : 736 × 1308

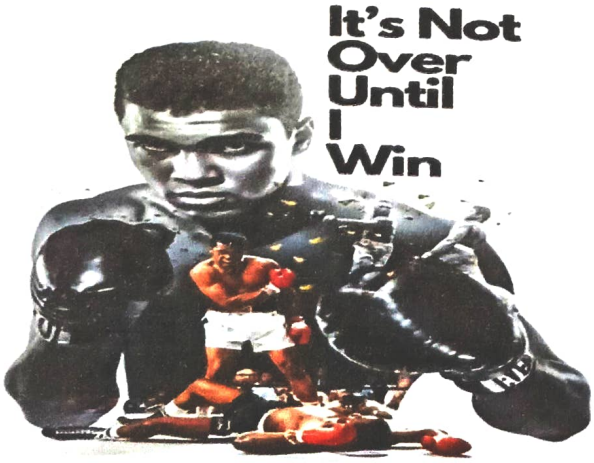
Number of Channels: 3

Data Type : uint8

Total Pixels : 962,688

Total Values : 2888064

Original Color Image (RGB)



✅ Image successfully loaded using OpenCV.

✅ OpenCV internally represents color images in BGR format.

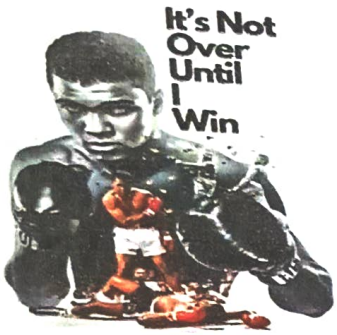
FILE FORMAT COMPARISON

JPEG File Size : 286.96 KB

PNG File Size : 1609.12 KB

Color Space Conversion

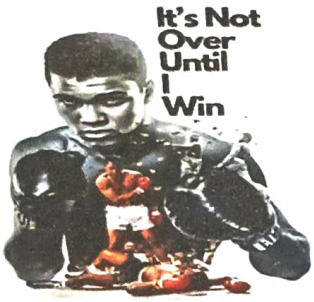
Original RGB Image



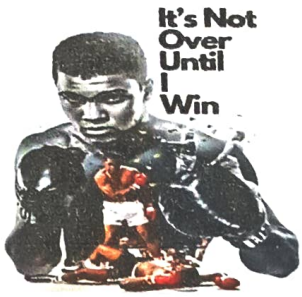
Grayscale



HSV Color Space



LAB Color Space



ROI INFORMATION

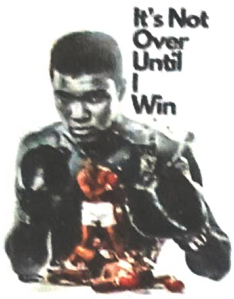
ROI Width : 441 pixels

ROI Height : 784 pixels

ROI Coordinates : (147, 262) to (588, 1046)

Geometric Transformations and ROI

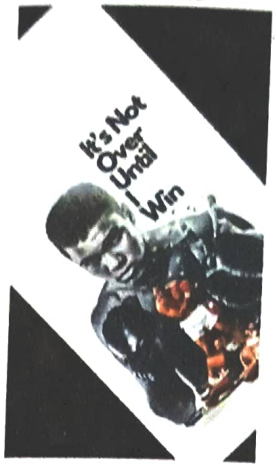
Original Image



Resized - 50%



Rotation - 45°



Horizontal Flip



Vertical Flip



Central ROI

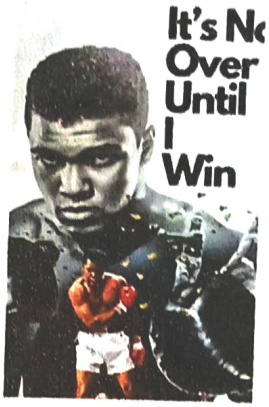


Image Complement Operation

Original Image

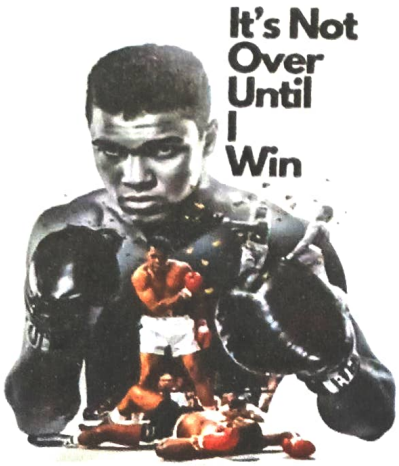
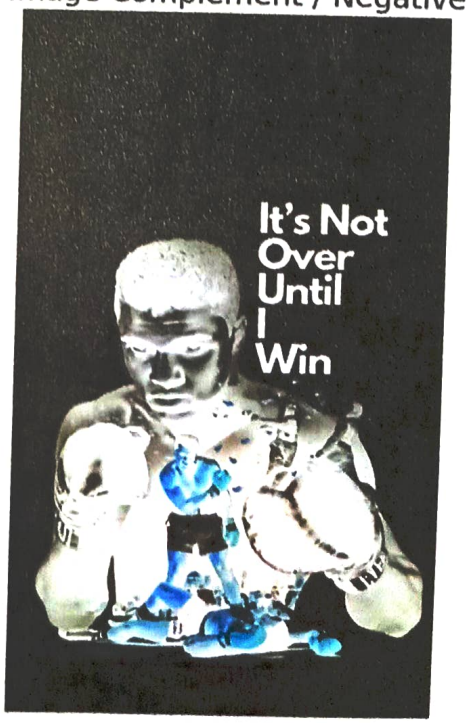
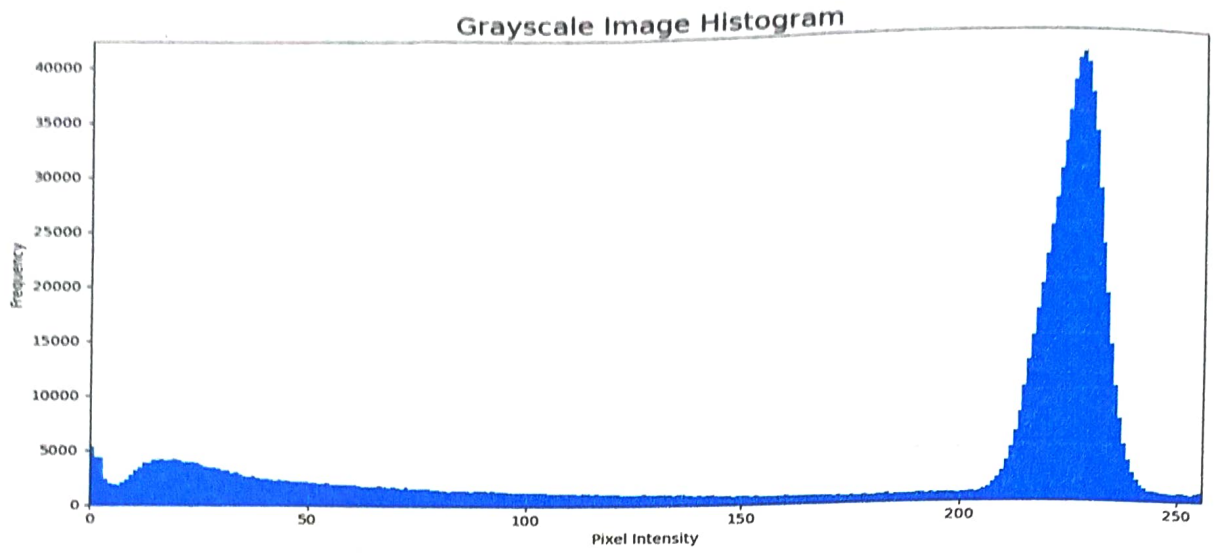


Image Complement / Negative



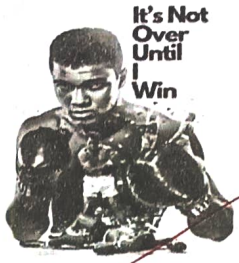


Fundamental Image Processing Operations

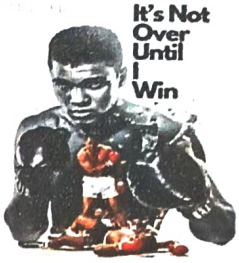
1. Original



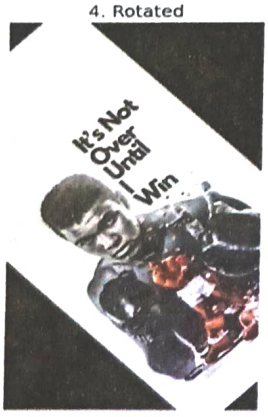
2. Grayscale



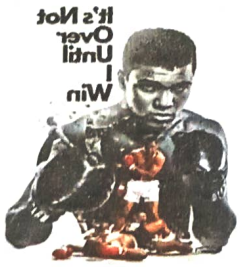
3. Resized



4. Rotated



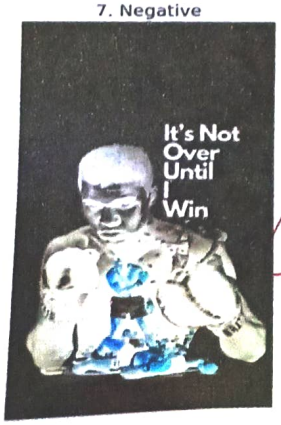
5. Horizontal Flip



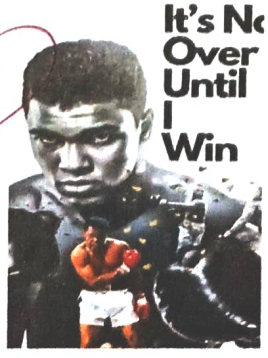
6. Vertical Flip



7. Negative



8. ROI



9. LAB Visualization

